

# Network Administration / System Administration

## Homework 2

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### Discussion Partners:

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## 0. VM Setup

### Reference:

- [unable to retrieve community.db: 404 errors \(Arch Linux Forums\)](#)
- [sudo pacman -Syu fails with error: failed to commit transaction \(Stack Exchange\)](#)
- [OpenSSH \(ArchWiki\)](#)
- [Enable SSH on Arch Linux](#)

為了方便連線，我打算在 VM 當中配置 ssh server，使我可以利用 ssh 的方式進入虛擬機。步驟如下：

1. 執行 `run_vm.sh`，以 vnc 登入 VM 當中的 root 帳號。
2. 更新 archlinux 失敗，經查閱資料後，發現需要將 `/etc/pacman.conf` 中的 `[[community]]` 註解掉。
3. 更新 archlinux keyring

---

```
1 pacman -Sy archlinux-keyring
```

---

4. 更新 archlinux 並安裝 openssh：

---

```
1 pacman -Syu
2 pacman -S openssh
```

---

5. 啟動 sshd

---

```
1 systemctl enable sshd
2 systemctl start sshd
```

---

6. 離開 VM，並在 host 端修改 `run_vm.sh`，加入最後一行，使 host 端的 4321 port 轉發到 VM 的 22 port。

```
1  #!/bin/bash
2
3  echo "Starting VM at port $1"
4  echo "parameter is $((($1 - 5900))"
5
6  qemu-system-x86_64 -m 2G --enable-kvm \
7      -vnc :"$((($1 - 5900))",password=on -monitor stdio \
8      -drive file=disk1.qcow2,media=disk,index=0 \
9      -drive file=disk2.qcow2,if=virtio,media=disk,index=1 \
10     -drive file=disk3.qcow2,if=virtio,media=disk,index=2 \
11     -drive file=disk4.qcow2,if=virtio,media=disk,index=3 \
12     -drive file=disk5.qcow2,if=virtio,media=disk,index=4 \
13     -drive file=disk6.qcow2,if=virtio,media=disk,index=5 \
14     -drive file=disk7.qcow2,if=virtio,media=disk,index=6 \
15     -drive file=disk8.qcow2,if=virtio,media=disk,index=7 \
16     -drive file=disk9.qcow2,if=virtio,media=disk,index=8 \
17     -drive file=disk10.qcow2,if=virtio,media=disk,index=9 \
18     -net user,hostfwd=tcp::4321-:22 -net nic
```

7. 成功從工作站當中以 balu 的帳號 ssh 進去 VM。

```
1  ssh balu@localhost -p 4321
```

## 1. 那傢伙竟然敢無視窗

### Reference:

- [NTFS3](#)
- [Journaling file system \(Wikipedia\)](#)
- [How to mount NTFS in Linux without fuseblk \(Stack Overflow\)](#)
- [使用 ntfs3 驱动替换 ntfs-3g 挂载 windows NTFS 分区](#)
- [对 NTFS 硬盘进行读写操作；开机自动挂载硬盘 \(CSDN\)](#)

NTFS 檔案系統可以滿足題目所要求，在雙系統中支援原生讀寫，並且在 Windows 11 下支援日誌功能。為了格式化 NTFS 檔案系統，需要先安裝 `ntfs-3g` 套件。我完成此題的步驟如下：

1. 在 VM 當中安裝 `ntfs-3g` 套件

---

```
1 sudo pacman -S ntfs-3g
```

---

## 2. 創立掛載點

---

```
1 sudo mkdir /mnt/myusb
```

---

## 3. 格式化硬碟

---

```
1 sudo mkfs.ntfs /dev/vdi2
```

---

## 4. 取得 UUID

---

```
1 sudo blkid /dev/vdi2
```

---

```
[balu@archlinux ~]$ sudo blkid /dev/vdi2
/dev/vdi2: BLOCK_SIZE="512" UUID="4E800CA236A749BE" TYPE="ntfs" PARTUUID="eb45fe31-f8c2-fd42-af97-b614bbf57e9b"
[balu@archlinux ~]$
```

Figure 1: Screenshot of `sudo blkid /dev/vdi2`

## 5. 配置自動掛載硬碟: 修改 `/etc/fstab`

## 6. 重開機，即可看到硬碟已經掛載

*Ans.*

```
[balu@archlinux ~]$ cat /etc/fstab
# Static information about the filesystems.
# See fstab(5) for details.

# <file system> <dir> <type> <options> <dump> <pass>
# /dev/sda2
UUID=d1daff5a-54da-43b8-a88e-83fa4e94a0b1 / ext4 rw,relatime 0 1

# /dev/sda1
UUID=711c-6167 /boot vfat rw,relatime,fmask=0022,dmask=0022,codepage=437,iocharset=ascii,shortname=m
ixed,utf8,errors=remount-ro 0 2

# /dev/vdi2
UUID=4E800CA236A749BE /mnt/myusb ntfs3 defaults 0 0

/dev/nasahw2-main/course /home/balu/course ext4 defaults 0 2
/dev/nasahw2-secondary/videos /home/balu/videos ext4 defaults 0 2
[balu@archlinux ~]$
```

Figure 2: Screenshot of `cat /etc/fstab`

```

[balu@archlinux ~]$ lsblk; df -hT
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
fd0                                2:0      1    4K  0 disk
sda                                8:0      0   32G  0 disk
├─sda1                             8:1      0  200M  0 part /boot
├─sda2                             8:2      0  31.8G  0 part /
sr0                               11:0     1 1024M  0 rom
zram0                             253:0    0   983M  0 disk [SWAP]
vda                               254:0    0    1G  0 disk
├─vda1                             254:1    0 1022M  0 part
├─┌nasahw2--main-course            252:0    0   500M  0 lvm  /home/balu/course
└─vdb                             254:16   0    1G  0 disk
├─vdb1                             254:17   0 1022M  0 part
├─vdc                             254:32   0    2G  0 disk
├─vdc1                             254:33   0    2G  0 part
├─vdd                             254:48   0   16G  0 disk
├─vdd1                             254:49   0   16G  0 part
├─vde                             254:64   0   512M  0 disk
├─vde1                             254:65   0   510M  0 part
├─┌nasahw2--secondary-videos      252:1    0   508M  0 lvm  /home/balu/videos
└─vdf                             254:80   0    7G  0 disk
├─vdf1                             254:81   0    2G  0 part
├─vdf2                             254:82   0    2G  0 part
├─vdf3                             254:83   0    2G  0 part
├─vdg                             254:96   0    7G  0 disk
├─vdg1                             254:97   0    2G  0 part
├─vdg2                             254:98   0    2G  0 part
├─vdg3                             254:99   0    2G  0 part
├─vdh                             254:112  0    7G  0 disk
├─vdh1                             254:113  0    2G  0 part
├─vdh2                             254:114  0    2G  0 part
├─vdh3                             254:115  0    2G  0 part
├─vdi                             254:128  0    6G  0 disk
├─vdi1                             254:129  0    2G  0 part
└─vdi2                             254:130  0    4G  0 part /mnt/myusb
Filesystem                        Type      Size  Used Avail Use% Mounted on
dev                              devtmpfs  976M    0  976M   0% /dev
run                              tmpfs     984M  708K  983M   1% /run
/dev/sda2                        ext4      32G   2.6G   27G   9% /
tmpfs                            tmpfs     984M    0  984M   0% /dev/shm
tmpfs                            tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs                            tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-resolved.service
tmpfs                            tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-networkd.service
tmpfs                            tmpfs     984M    0  984M   0% /tmp
/dev/vdi2                        ntfs3     4.0G   22M   4.0G   1% /mnt/myusb
/dev/sda1                        vfat      197M   69M  129M  35% /boot
/dev/mapper/nasahw2--main-course ext4      459M   4.5M  425M   2% /home/balu/course
/dev/mapper/nasahw2--secondary-videos ext4     466M   66M  371M  16% /home/balu/videos
tmpfs                            tmpfs     1.0M    0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs                            tmpfs     197M   4.0K  197M   1% /run/user/1000
[balu@archlinux ~]$

```

Figure 3: Screenshot of lsblk; df -hT

## 2. 因為要換到新的 SWAP

### Reference:

- [Create a Linux Swap File \(Linuxize\)](#)

新增 SWAP 檔案的步驟如下：

1. 建立 /newswap 檔案

```
1 sudo fallocate -l 4G /newswap
```

2. 設定檔案權限

```
1 sudo chmod 600 /newswap
```

3. 格式化檔案

```
1 sudo mkswap /newswap
```

4. 啟用 SWAP 檔案

```
1 sudo swapon /newswap
```

*Ans.*



```
[balu@archlinux ~]$ free -h
              total        used         free       shared    buff/cache   available
Mem:           1.9Gi       207Mi       1.7Gi         4.6Mi        103Mi        1.7Gi
Swap:          5.0Gi           0B         5.0Gi
```

Figure 4: Screenshot of `free -h`



### 3. 為資料創造新的棲身之處

#### Reference:

- [lvextend 命令 \(LinuxCool\)](#)

使用指令如下：

```
1 sudo lvextend -L 1GiB /dev/nasahw2-main/course
```

Ans.

```
[root@archlinux ~]# lsblk; df -hT
```

| NAME                          | MAJ:MIN | RM | SIZE  | RO | TYPE | MOUNTPOINTS       |
|-------------------------------|---------|----|-------|----|------|-------------------|
| fd0                           | 2:0     | 1  | 4K    | 0  | disk |                   |
| sda                           | 8:0     | 0  | 32G   | 0  | disk |                   |
| └─sda1                        | 8:1     | 0  | 200M  | 0  | part | /boot             |
| └─sda2                        | 8:2     | 0  | 31.8G | 0  | part | /                 |
| sr0                           | 11:0    | 1  | 1024M | 0  | rom  |                   |
| zram0                         | 253:0   | 0  | 983M  | 0  | disk | [SWAP]            |
| vda                           | 254:0   | 0  | 1G    | 0  | disk |                   |
| └─vda1                        | 254:1   | 0  | 1022M | 0  | part |                   |
| └─┬─nasahw2--main-course      | 252:0   | 0  | 1G    | 0  | lvm  | /home/balu/course |
| vdb                           | 254:16  | 0  | 1G    | 0  | disk |                   |
| └─vdb1                        | 254:17  | 0  | 1022M | 0  | part |                   |
| └─┬─nasahw2--main-course      | 252:0   | 0  | 1G    | 0  | lvm  | /home/balu/course |
| vdc                           | 254:32  | 0  | 2G    | 0  | disk |                   |
| └─vdc1                        | 254:33  | 0  | 2G    | 0  | part |                   |
| vdd                           | 254:48  | 0  | 16G   | 0  | disk |                   |
| └─vdd1                        | 254:49  | 0  | 16G   | 0  | part |                   |
| vde                           | 254:64  | 0  | 512M  | 0  | disk |                   |
| └─vde1                        | 254:65  | 0  | 510M  | 0  | part |                   |
| └─┬─nasahw2--secondary-videos | 252:1   | 0  | 508M  | 0  | lvm  | /home/balu/videos |
| vdf                           | 254:80  | 0  | 7G    | 0  | disk |                   |
| └─vdf1                        | 254:81  | 0  | 2G    | 0  | part |                   |
| └─vdf2                        | 254:82  | 0  | 2G    | 0  | part |                   |
| └─vdf3                        | 254:83  | 0  | 2G    | 0  | part |                   |
| vdg                           | 254:96  | 0  | 7G    | 0  | disk |                   |
| └─vdg1                        | 254:97  | 0  | 2G    | 0  | part |                   |
| └─vdg2                        | 254:98  | 0  | 2G    | 0  | part |                   |
| └─vdg3                        | 254:99  | 0  | 2G    | 0  | part |                   |
| vdh                           | 254:112 | 0  | 7G    | 0  | disk |                   |
| └─vdh1                        | 254:113 | 0  | 2G    | 0  | part |                   |
| └─vdh2                        | 254:114 | 0  | 2G    | 0  | part |                   |
| └─vdh3                        | 254:115 | 0  | 2G    | 0  | part |                   |
| vdi                           | 254:128 | 0  | 6G    | 0  | disk |                   |
| └─vdi1                        | 254:129 | 0  | 2G    | 0  | part |                   |
| └─vdi2                        | 254:130 | 0  | 4G    | 0  | part | /mnt/myusb        |

  

| Filesystem                            | Type     | Size | Used | Avail | Use% | Mounted on                                |
|---------------------------------------|----------|------|------|-------|------|-------------------------------------------|
| dev                                   | devtmpfs | 976M | 0    | 976M  | 0%   | /dev                                      |
| run                                   | tmpfs    | 984M | 712K | 983M  | 1%   | /run                                      |
| /dev/sda2                             | ext4     | 32G  | 6.6G | 23G   | 23%  | /                                         |
| tmpfs                                 | tmpfs    | 984M | 0    | 984M  | 0%   | /dev/shm                                  |
| tmpfs                                 | tmpfs    | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/systemd-journald.service |
| tmpfs                                 | tmpfs    | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/systemd-resolved.service |
| tmpfs                                 | tmpfs    | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/systemd-networkd.service |
| tmpfs                                 | tmpfs    | 984M | 0    | 984M  | 0%   | /tmp                                      |
| /dev/vdi2                             | ntfs3    | 4.0G | 22M  | 4.0G  | 1%   | /mnt/myusb                                |
| /dev/sda1                             | vfat     | 197M | 69M  | 129M  | 35%  | /boot                                     |
| /dev/mapper/nasahw2--main-course      | ext4     | 459M | 4.5M | 425M  | 2%   | /home/balu/course                         |
| /dev/mapper/nasahw2--secondary-videos | ext4     | 466M | 66M  | 371M  | 16%  | /home/balu/videos                         |
| tmpfs                                 | tmpfs    | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/getty@tty1.service       |
| tmpfs                                 | tmpfs    | 197M | 4.0K | 197M  | 1%   | /run/user/1000                            |

```
[root@archlinux ~]#
```

Figure 5: Screenshot of `lsblk; df -hT`

## 4. 我有拜託妳別把我的作業告訴其他人了吧

### Reference:

- [How to Encrypt a Drive in Linux \(Medium\)](#)
- [How to add a passphrase, key, or keyfile to an existing LUKS device \(Red Hat\)](#)
- [Linux 使用 LUKS 加密 Ext4 檔案系統](#)

我做的步驟如下：

1. 以 root 進入 VM。

2. 切出一塊新的 lvm，作為加密區域

```
1 lvcreate -L 800MiB -n homework nasahw2-main
```

3. 加密 /dev/nasahw2-main/homework

```
1 cryptsetup luksFormat /dev/nasahw2-main/homework
```

4. 開啟加密區域

```
1 cryptsetup open /dev/nasahw2-main/homework homework
```

5. 格式化加密區域

```
1 mkfs.ext4 /dev/mapper/homework
```

6. 掛載加密區域

```
1 mount /dev/mapper/homework /home/balu/homework/
```

7. 設定 keyfile

```
1 cryptsetup luksAddKey /dev/nasahw2-main/homework /home/balu/lvm_key
```

8. 設定開機自動解密，在 /etc/crypttab 中加入底下這行：

```
1 homework /dev/nasahw2-main/homework /home/balu/lvm_key
```

9. 設定開機自動 mount，在 /etc/fstab 中加入底下這行：

```
1 /dev/mapper/homework /home/balu/homework ext4 defaults 0 2
```

## 10. 重開機，確認加密區域已經掛載

Ans.

```
[root@archlinux balu]# lsblk; df -hT
```

| NAME                        | MAJ:MIN | RM | SIZE  | RO | TYPE  | MOUNTPOINTS         |
|-----------------------------|---------|----|-------|----|-------|---------------------|
| fd0                         | 2:0     | 1  | 4K    | 0  | disk  |                     |
| sda                         | 8:0     | 0  | 32G   | 0  | disk  |                     |
| └sda1                       | 8:1     | 0  | 200M  | 0  | part  | /boot               |
| └sda2                       | 8:2     | 0  | 31.8G | 0  | part  | /                   |
| sr0                         | 11:0    | 1  | 1024M | 0  | rom   |                     |
| zram0                       | 253:0   | 0  | 983M  | 0  | disk  | [SWAP]              |
| vda                         | 254:0   | 0  | 1G    | 0  | disk  |                     |
| └vda1                       | 254:1   | 0  | 1022M | 0  | part  |                     |
| └└nasahw2--main-course      | 252:0   | 0  | 1G    | 0  | lvm   | /home/balu/course   |
| vdb                         | 254:16  | 0  | 1G    | 0  | disk  |                     |
| └vdb1                       | 254:17  | 0  | 1022M | 0  | part  |                     |
| └└nasahw2--main-course      | 252:0   | 0  | 1G    | 0  | lvm   | /home/balu/course   |
| └└nasahw2--main-homework    | 252:1   | 0  | 800M  | 0  | lvm   |                     |
| └└└homework                 | 252:3   | 0  | 784M  | 0  | crypt | /home/balu/homework |
| vdc                         | 254:32  | 0  | 2G    | 0  | disk  |                     |
| └vdc1                       | 254:33  | 0  | 2G    | 0  | part  |                     |
| vdd                         | 254:48  | 0  | 16G   | 0  | disk  |                     |
| └vdd1                       | 254:49  | 0  | 16G   | 0  | part  |                     |
| vde                         | 254:64  | 0  | 512M  | 0  | disk  |                     |
| └vde1                       | 254:65  | 0  | 510M  | 0  | part  |                     |
| └└nasahw2--secondary-videos | 252:2   | 0  | 508M  | 0  | lvm   | /home/balu/videos   |
| vdf                         | 254:80  | 0  | 7G    | 0  | disk  |                     |
| └vdf1                       | 254:81  | 0  | 2G    | 0  | part  |                     |
| └vdf2                       | 254:82  | 0  | 2G    | 0  | part  |                     |
| └vdf3                       | 254:83  | 0  | 2G    | 0  | part  |                     |
| vdg                         | 254:96  | 0  | 7G    | 0  | disk  |                     |
| └vdg1                       | 254:97  | 0  | 2G    | 0  | part  |                     |
| └vdg2                       | 254:98  | 0  | 2G    | 0  | part  |                     |
| └vdg3                       | 254:99  | 0  | 2G    | 0  | part  |                     |
| vdh                         | 254:112 | 0  | 7G    | 0  | disk  |                     |
| └vdh1                       | 254:113 | 0  | 2G    | 0  | part  |                     |
| └vdh2                       | 254:114 | 0  | 2G    | 0  | part  |                     |
| └vdh3                       | 254:115 | 0  | 2G    | 0  | part  |                     |
| vdi                         | 254:128 | 0  | 6G    | 0  | disk  |                     |
| └vdi1                       | 254:129 | 0  | 2G    | 0  | part  |                     |
| └vdi2                       | 254:130 | 0  | 4G    | 0  | part  | /mnt/myusb          |

  

| Filesystem                            | Type     | Size | Used | Avail | Use% | Mounted on                                           |
|---------------------------------------|----------|------|------|-------|------|------------------------------------------------------|
| dev                                   | devtmpfs | 976M | 0    | 976M  | 0%   | /dev                                                 |
| run                                   | tmpfs    | 984M | 732K | 983M  | 1%   | /run                                                 |
| /dev/sda2                             | ext4     | 32G  | 6.6G | 23G   | 23%  | /                                                    |
| tmpfs                                 | tmpfs    | 984M | 0    | 984M  | 0%   | /dev/shm                                             |
| tmpfs                                 | tmpfs    | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/systemd-journald.service            |
| tmpfs                                 | tmpfs    | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/systemd-resolved.service            |
| tmpfs                                 | tmpfs    | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/systemd-networkd.service            |
| tmpfs                                 | tmpfs    | 984M | 0    | 984M  | 0%   | /tmp                                                 |
| /dev/vdi2                             | ntfs3    | 4.0G | 22M  | 4.0G  | 1%   | /mnt/myusb                                           |
| /dev/sda1                             | vfat     | 197M | 69M  | 129M  | 35%  | /boot                                                |
| tmpfs                                 | tmpfs    | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/systemd-cryptsetup@homework.service |
| /dev/mapper/nasahw2--main-course      | ext4     | 459M | 4.5M | 425M  | 2%   | /home/balu/course                                    |
| /dev/mapper/nasahw2--secondary-videos | ext4     | 466M | 66M  | 371M  | 16%  | /home/balu/videos                                    |
| /dev/mapper/homework                  | ext4     | 755M | 220K | 700M  | 1%   | /home/balu/homework                                  |
| tmpfs                                 | tmpfs    | 1.0M | 0    | 1.0M  | 0%   | /run/credentials/getty@tty1.service                  |
| tmpfs                                 | tmpfs    | 197M | 4.0K | 197M  | 1%   | /run/user/1000                                       |

```
[root@archlinux balu]#
```

Figure 6: Screenshot of lsblk; df -hT



## 5. 快照真的好難喔

### Reference:

- [LVM 系統快照設定教學 \(iThome\)](#)
- [Linux 解压缩 tar.zst 类型文件 \(CSDN\)](#)
- [How to keep or drop LVM snapshot \(Server Fault\)](#)

我以 root 進入 VM，並進行以下操作：

1. 將 /dev/vdc1 設為 pv，並且加到 nasahw2-main vg 中

```
1 pvcreate /dev/vdc1
2 vgextend nasahw2-main /dev/vdc1
```

2. 建立 snapshot 的 lv

```
1 lvcreate -L 1GiB -s -n backup /dev/nasahw2-main/course
```

3. 掛載 /dev/nasahw2-main/backup 到 /mnt/backup

```
1 mkdir -p /mnt/backup
2 mount /dev/nasahw2-main/backup /mnt/backup
```

4. lsblk 結果如 Figure 7 所示。

5. 進行備份到 tar:

```
1 tar -I zstd -cvf /home/balu/backup.tar.zst /mnt/backup
```

6. 卸載 /mnt/backup，並刪除 snapshot。

```
1 umount /mnt/backup
2 lvremove /dev/nasahw2-main/backup
```

7. 執行 lsblk; df -hT，結果如 Figure 8 所示。

```

[root@archlinux balu]# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
fd0                                 2:0      1    4K  0 disk
sda                                 8:0      0   32G  0 disk
├─sda1                             8:1      0   200M  0 part  /boot
└─sda2                             8:2      0  31.8G  0 part  /
sr0                                 11:0     1  1024M  0 rom
zram0                             253:0     0    983M  0 disk  [SWAP]
vda                                 254:0     0     1G  0 disk
├─vda1                             254:1     0  1022M  0 part
│   └─nasahw2--main-course-real 252:4     0     1G  0 lvm
│       ├──nasahw2--main-course 252:0     0     1G  0 lvm  /home/balu/course
│       └─nasahw2--main-backup 252:6     0     1G  0 lvm  /mnt/backup
└─vdb                               254:16    0     1G  0 disk
    ├─vdb1                         254:17    0  1022M  0 part
    │   └─nasahw2--main-homework 252:1     0   800M  0 lvm
    │       └─homework           252:3     0   784M  0 crypt /home/balu/homework
    └─nasahw2--main-course-real 252:4     0     1G  0 lvm
        ├──nasahw2--main-course 252:0     0     1G  0 lvm  /home/balu/course
        └─nasahw2--main-backup 252:6     0     1G  0 lvm  /mnt/backup
vdc                                 254:32    0     2G  0 disk
├─vdc1                             254:33    0     2G  0 part
│   └─nasahw2--main-backup-cow 252:5     0     1G  0 lvm
│       └─nasahw2--main-backup 252:6     0     1G  0 lvm  /mnt/backup
vdd                                 254:48    0    16G  0 disk
└─vdd1                             254:49    0    16G  0 part
vde                                 254:64    0    512M  0 disk
├─vde1                             254:65    0    510M  0 part
│   └─nasahw2--secondary-videos 252:2     0    508M  0 lvm  /home/balu/videos
└─vdf                               254:80    0     7G  0 disk
    ├──vdf1                         254:81    0     2G  0 part
    ├──vdf2                         254:82    0     2G  0 part
    └─vdf3                         254:83    0     2G  0 part
vdg                                 254:96    0     7G  0 disk
├─vdg1                             254:97    0     2G  0 part
├─vdg2                             254:98    0     2G  0 part
└─vdg3                             254:99    0     2G  0 part
vdh                                 254:112   0     7G  0 disk
├─vdh1                             254:113   0     2G  0 part
├─vdh2                             254:114   0     2G  0 part
└─vdh3                             254:115   0     2G  0 part
vdi                                 254:128   0     6G  0 disk
├─vdi1                             254:129   0     2G  0 part
└─vdi2                             254:130   0     4G  0 part  /mnt/myusb
[root@archlinux balu]#

```

Figure 7: Screenshot of lsblk after creating snapshot

```

[root@archlinux balu]# lsblk; df -hT
NAME                                MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
fd0                                2:0      1   4K  0 disk
sda                                8:0      0  32G  0 disk
├─sda1                             8:1      0 200M  0 part  /boot
└─sda2                             8:2      0 31.8G  0 part  /
sr0                               11:0      1 1024M  0 rom
zram0                             253:0      0 983M  0 disk  [SWAP]
vda                               254:0      0   1G  0 disk
├─vda1                             254:1      0 1022M  0 part
└─nasahw2--main-course             252:0      0   1G  0 lvm    /home/balu/course
vdb                               254:16     0   1G  0 disk
├─vdb1                             254:17     0 1022M  0 part
└─nasahw2--main-course             252:0      0   1G  0 lvm    /home/balu/course
    └─nasahw2--main-homework       252:1      0 800M  0 lvm
        └─homework                 252:3      0 784M  0 crypt /home/balu/homework
vdc                               254:32     0   2G  0 disk
├─vdc1                             254:33     0   2G  0 part
vdd                               254:48     0  16G  0 disk
├─vdd1                             254:49     0  16G  0 part
vde                               254:64     0  512M  0 disk
├─vde1                             254:65     0  510M  0 part
└─nasahw2--secondary-videos       252:2      0 508M  0 lvm    /home/balu/videos
vdf                               254:80     0   7G  0 disk
├─vdf1                             254:81     0   2G  0 part
├─vdf2                             254:82     0   2G  0 part
└─vdf3                             254:83     0   2G  0 part
vdg                               254:96     0   7G  0 disk
├─vdg1                             254:97     0   2G  0 part
├─vdg2                             254:98     0   2G  0 part
└─vdg3                             254:99     0   2G  0 part
vdh                               254:112    0   7G  0 disk
├─vdh1                             254:113    0   2G  0 part
├─vdh2                             254:114    0   2G  0 part
└─vdh3                             254:115    0   2G  0 part
vdi                               254:128    0   6G  0 disk
├─vdi1                             254:129    0   2G  0 part
└─vdi2                             254:130    0   4G  0 part  /mnt/myusb
Filesystem                        Type      Size  Used Avail Use% Mounted on
dev                              devtmpfs  976M    0  976M   0% /dev
run                              tmpfs     984M   740K  983M   1% /run
/dev/sda2                       ext4      32G    6.6G   23G  23% /
tmpfs                           tmpfs     984M    0  984M   0% /dev/shm
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-resolved.service
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-networkd.service
tmpfs                           tmpfs     984M    0  984M   0% /tmp
/dev/vdi2                       ntfs3     4.0G   22M   4.0G   1% /mnt/myusb
/dev/sda1                       vfat      197M    69M  129M  35% /boot
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/systemd-cryptsetup@homework.service
/dev/mapper/nasahw2--main-course ext4      459M   4.5M  425M   2% /home/balu/course
/dev/mapper/nasahw2--secondary-videos ext4      466M    66M  371M  16% /home/balu/videos
/dev/mapper/homework            ext4      755M  220K   700M   1% /home/balu/homework
tmpfs                           tmpfs     1.0M    0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs                           tmpfs     197M   4.0K  197M   1% /run/user/1000

```

Figure 8: Screenshot of `lsblk; df -hT` after removing snapshot

## 6. 好老舊喔

## 7. 我看還是再來合一次吧

8. 等一下，妳還沒回答我